Polarized Characterization Interface

1 Polarized Characterization Interface

This section records the finite-cutoff characterization data extracted from the split/leak construction and used in the 2T Main assumption package Definition 4.2. In this note, an "accepted tick" denotes the local accepted-branch slice at cutoff Λ_k.

For each accepted tick k, write

$$a_{\Lambda_k} := a_k, \qquad S_{\Lambda_k} := a_{\Lambda_k}^2.$$

The variable a_{Λ_k} is the distinguished odd split coordinate identified in sections 1.1 and 1.3; the even quantity S_{Λ_k} is the positive stock that enters the leak law and the split threshold.

Definition 1.1 (Characterization algebra, operational triviality, and even null ideal)**.** Fix an accepted tick k and let $E^-_{\Lambda_k}$ denote the physically odd sufficiently local generator space at cutoff Λ_k, i.e. the odd sector for the transported physical involution Γ. Let $E^+_{\Lambda_k}$ denote the corresponding physically even sufficiently local observable space. Define the core characterization algebra by

$$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} := \mathrm{alg}\{a_{\Lambda_k},\, S_{\Lambda_k}\}. \tag{1}$$

Define the operationally interface-trivial odd subspace by

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} := \overline{\Big\{T \in E^-_{\Lambda_k} : \{T, O\}_{\mathrm{P},\Lambda_k} = 0 \ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\Big\}}. \tag{2}$$

Define the even interface-null ideal by

$$\mathcal{N}^{\mathrm{even}}_{\Lambda_k} := \overline{\Big\{E \in E^+_{\Lambda_k} : \{E, O\}_{\mathrm{P},\Lambda_k} = 0 \ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\Big\}}. \tag{3}$$

Remark 1.2 (Do not identify the characterization observable with the generator)*.* The odd observable a_{Λ_k} characterizes the active completion-sector channel, but it is not itself the odd Hamiltonian generator class appearing in item **(M4)** of Definition 4.2. That generator is denoted $J^{\mathrm{act}}_{\Lambda_k} \in E^-_{\Lambda_k}$ below. The quotient statement belongs to $[J^{\mathrm{act}}_{\Lambda_k}]$, not to $[a_{\Lambda_k}]$.

Definition 1.3 (Intrinsic covariance-tangent response map)**.** Fix an accepted tick k. Let $\mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ denote the finite-cutoff receiver covariance-tangent space at the UOM reference Gaussian state ρ_{*,Λ_k}, equipped with the Fisher pairing

$$\langle\!\langle \delta C, \delta C' \rangle\!\rangle_{C_{*,\Lambda_k}} := \tfrac{1}{2}\,\mathrm{tr}\big(C^{-1}_{*,\Lambda_k}\delta C\, C^{-1}_{*,\Lambda_k}\delta C'\big).$$

For admissible band-limited receiver probes f, g, let

$$\mathcal{Q}^{\Lambda_k}_{f,g} \in E^+_{\Lambda_k}$$

denote the corresponding sufficiently local quadratic observable, normalized so that on centered Gaussian states with covariance C,

$$\left\langle \mathcal{Q}^{\Lambda_k}_{f,g} \right\rangle_{\rho(C)} = \langle f, Cg \rangle.$$

For $F \in E^-_{\Lambda_k}$, define $\mathcal{R}_{\Lambda_k}(F) \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ by the matrix elements

$$\left\langle f, \mathcal{R}_{\Lambda_k}(F) g \right\rangle := \left\langle \{F, \mathcal{Q}^{\Lambda_k}_{f,g}\}_{\mathrm{P},\Lambda_k} \right\rangle_{\rho_{*,\Lambda_k}}.$$

We call

$$\mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$$

the *intrinsic covariance-tangent response map.*

Proposition 1.4 (Intrinsic definition and completion independence)**.** *The map $\mathcal{R}_{\Lambda_k}$ of definition 1.3 is continuous and linear. It depends only on the finite-cutoff UOM Peierls package and the UOM covariance/Fisher package. If a Gaussian completion realization is chosen, then $\mathcal{R}_{\Lambda_k}(F)$ is represented by the same completion-sector covariance tangent that one obtains by linearizing the completion quadratures. Thus Gaussian completion is a realization of the response map, not extra structure required to define it.*

Proof. For fixed $F \in E^-_{\Lambda_k}$, the finite-cutoff UOM Peierls package makes

$$\delta_F := \{F, -\}_{\mathrm{P},\Lambda_k}$$

a continuous derivation on the sufficiently local algebra. Applying δ_F to the quadratic probes $\mathcal{Q}^{\Lambda_k}_{f,g}$ and evaluating in the reference Gaussian state produces a continuous symmetric bilinear form in f, g. In the UOM Gaussian radiative sector, such bilinear forms are exactly covariance tangents at C_{*,Λ_k}, equipped with the Fisher metric. Linearity and continuity in F are inherited directly from the Peierls bracket. No completion coordinates are used in this construction.

If a Gaussian completion is later chosen, the same second moments are encoded by the covariance matrix of the completion quadratures. The bilinear form defined above and the completion-sector covariance tangent therefore represent the same infinitesimal response object in two different coordinate systems. □

Definition 1.5 (Physical intrinsic covariance-response map)**.** Fix an accepted tick k. Let

$$\mathcal{T}^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathcal{T}^{\mathrm{cov}}_{\Lambda_k} \subset \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}.$$

Define the physical intrinsic covariance-response map by

$$R^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}.$$

Proposition 1.6 (Physical intrinsic response map)**.** *The map $R^{\mathrm{phys}}_{\Lambda_k}$ of definition 1.5 is continuous and linear.*

Proof. This is immediate from proposition 1.4 and the continuity and linearity of the finite-cutoff physical projector $\Pi_{\mathrm{phys},\Lambda_k}$. □

Proposition 1.7 (Reduction of operational triviality to the active odd coordinate)**.** *Fix an accepted tick k. If $F \in E^-_{\Lambda_k}$ satisfies*

$$\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

then $F \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. Since $S_{\Lambda_k} = a_{\Lambda_k}^2$ and the Peierls bracket is a derivation,

$$\{F, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = \{F, a_{\Lambda_k}^2\}_{\mathrm{P},\Lambda_k} = 2\, a_{\Lambda_k}\, \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

The algebra $\mathcal{O}_{\Lambda_k}^{\mathrm{char}} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ is generated by a_{Λ_k} and S_{Λ_k}, and the Peierls bracket acts on it by the Leibniz rule. Hence $\{F, O\}_{\mathrm{P},\Lambda_k} = 0$ for every $O \in \mathcal{O}_{\Lambda_k}^{\mathrm{char}}$, proving $F \in \mathcal{T}_{\Lambda_k}^{\mathrm{op}}$. □

Assumption 1.1 (Active odd generator and intrinsic-response factorization)**.** For each accepted tick k, there exist a distinguished active odd generator

$$J_{\Lambda_k}^{\mathrm{act}} \in E_{\Lambda_k}^{-}$$

and a nonzero physical covariance tangent

$$C_{\Lambda_k}^{\mathrm{act}} \in \mathcal{T}_{\Lambda_k}^{\mathrm{phys}}$$

and a continuous nonzero linear functional

$$\rho_{\Lambda_k} : E_{\Lambda_k}^{-} \to \mathbb{R}$$

normalized by

$$\rho_{\Lambda_k}(J_{\Lambda_k}^{\mathrm{act}}) = 1,$$

such that

$$R_{\Lambda_k}^{\mathrm{phys}}(J_{\Lambda_k}^{\mathrm{act}}) = C_{\Lambda_k}^{\mathrm{act}} \neq 0$$

and

$$R_{\Lambda_k}^{\mathrm{phys}}(F) = \rho_{\Lambda_k}(F)\, C_{\Lambda_k}^{\mathrm{act}} \qquad \forall F \in E_{\Lambda_k}^{-} \tag{4}$$

holds on the strong sector under consideration.

Remark 1.8 (Strong normalization)*.* If one further chooses the Fisher-unit normalization $\widehat{C}_{\Lambda_k}^{\mathrm{act}} := \|C_{\Lambda_k}^{\mathrm{act}}\|_{C_*,\Lambda_k}^{-1}\, C_{\Lambda_k}^{\mathrm{act}}$, then (4) sharpens to

$$R_{\Lambda_k}^{\mathrm{phys}}(F) = \widehat{\rho}_{\Lambda_k}(F)\, \widehat{C}_{\Lambda_k}^{\mathrm{act}}$$

with $\widehat{\rho}_{\Lambda_k} := \|C_{\Lambda_k}^{\mathrm{act}}\|_{C_*,\Lambda_k}\, \rho_{\Lambda_k}$. The export below does not require this extra normalization; it only needs the scalar response coefficient ρ_{Λ_k} relative to a fixed nonzero active tangent $C_{\Lambda_k}^{\mathrm{act}}$.

Assumption 1.2 (Physical-response sufficiency)**.** For each accepted tick k, the operationally trivial odd directions are exactly the kernel of the physical intrinsic covariance-response map:

$$\ker R_{\Lambda_k}^{\mathrm{phys}} = \mathcal{T}_{\Lambda_k}^{\mathrm{op}}.$$

Equivalently, an odd generator has zero physical intrinsic covariance response if and only if it is operationally trivial on the full core characterization algebra $\mathcal{O}_{\Lambda_k}^{\mathrm{char}}$.

Remark 1.9 (Why the strong package must be stated intrinsically)*.* The earlier quotient-valued formulation of the strong package in $E_{\Lambda_k}^{+}/\mathcal{N}_{\Lambda_k}^{\mathrm{even}}$ is not stable. Because the Peierls bracket here is the ordinary derivation bracket and $S_{\Lambda_k} = a_{\Lambda_k}^2$, the even observable $\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k}$ already Poisson-commutes with both a_{Λ_k} and S_{Λ_k}, hence lies automatically in $\mathcal{N}_{\Lambda_k}^{\mathrm{even}}$. The correct codomain for factorization and kernel identification is therefore the intrinsic physical covariance-response space $\mathcal{T}_{\Lambda_k}^{\mathrm{phys}}$, not the even observable quotient.

Proposition 1.10 (Operational triviality is the kernel of the active response)**.** *Assume assumptions 1.1 and 1.2. Then*

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} = \ker \rho_{\Lambda_k}.$$

Proof. Because $C^{\mathrm{act}}_{\Lambda_k} \neq 0$, the factorization identity (4) implies

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = 0 \qquad \Longleftrightarrow \qquad \rho_{\Lambda_k}(F) = 0$$

for every $F \in E^-_{\Lambda_k}$. Thus $\ker R^{\mathrm{phys}}_{\Lambda_k} = \ker \rho_{\Lambda_k}$. Apply assumption 1.2. □

Corollary 1.11 (Rank-one active odd quotient)**.** *Assume assumptions 1.1 and 1.2. Then*

$$E^-_{\Lambda_k}/\mathcal{T}^{\mathrm{op}}_{\Lambda_k} \cong \mathbb{R},$$

and the quotient is generated by the class of the active odd generator $[J^{\mathrm{act}}_{\Lambda_k}]$.

Proof. By proposition 1.10, the quotient is canonically isomorphic to the image of the nonzero functional ρ_{Λ_k}. Since ρ_{Λ_k} is real-valued and $\rho_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$, its image is $\mathbb{R}$ and the quotient is generated by $[J^{\mathrm{act}}_{\Lambda_k}]$. □

Proposition 1.12 (BRST/gauge-exact compatibility on the characterization layer)**.** *For each accepted tick k, the odd parts of the BRST/gauge-exact directions*

$$Q_t X + Q_u Y$$

lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. The exported characterization observables a_{Λ_k} and S_{Λ_k} are physical finite-cutoff observables on the characterization layer. The BRST/gauge-exact odd directions $Q_t X + Q_u Y$ represent redundancies of that physical layer, so their Peierls action vanishes on $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. By (2), their odd parts therefore lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Proposition 1.13 (Boundary-admissible divergence compatibility)**.** *Let K^μ be a sufficiently local current whose renormalized boundary flux vanishes on the core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. Then the odd part of*

$$\partial_\mu K^\mu$$

lies in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. For the admissible currents in the statement, the renormalized Stokes reduction of $\partial_\mu K^\mu$ produces no boundary term on the characterization layer. Hence its Peierls action on every $O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}$ vanishes, and the odd part belongs to $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by (2). □

Definition 1.14 (Admissible odd counterterms on the characterization layer)**.** For each accepted tick k, an odd sufficiently local counterterm ambiguity

$$\Delta^-_{\mathrm{ct}} \in E^-_{\Lambda_k}$$

is called *admissible on the characterization layer* if its intrinsic covariance response has no physical projection:

$$\Pi_{\mathrm{phys},\Lambda_k}\, \mathcal{R}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Equivalently, admissible odd counterterms are precisely those local scheme shifts whose response is confined to the RP, Weyl, conformal, or other even-null monitor sectors and does not enter the compressed active odd line.

Proposition 1.15 (Counterterm rigidity on the characterization layer)**.** *Assume that the characterization observable a_{Λ_k} is extracted from the physical covariance response so that*

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) = 0 \quad \Longrightarrow \quad \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0 \qquad \forall F \in E^-_{\Lambda_k}.$$

Then every admissible odd counterterm ambiguity in the sense of definition 1.14 acts trivially on the core characterization algebra:

$$\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Proof. If $F = \Delta^-_{\mathrm{ct}}$ is admissible, then this physical projection vanishes by definition. The displayed hypothesis therefore gives

$$\{\Delta^-_{\mathrm{ct}}, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

and since $S_{\Lambda_k} = a^2_{\Lambda_k}$, the derivation property yields

$$\{\Delta^-_{\mathrm{ct}}, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

By proposition 1.7, the first identity already implies $\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Remark 1.16 (Active stiffness versus full calibration)*.* The split/leak analysis determines the active stiffness $\widetilde{\alpha}(\Lambda_k)$ on the distinguished line generated by $J^{\mathrm{act}}_{\Lambda_k}$, with

$$\widetilde{\alpha}(\Lambda_k) \asymp \alpha(\Lambda_k) \sim \Lambda_k^{d+2}$$

by assumption 1.7. What UOM does *not* determine canonically is a positive bilinear form on the whole odd generator space $E^-_{\Lambda_k}$. That extension is auxiliary interface data, not a UOM-dynamical output.

Assumption 1.3 (Auxiliary coercive core on the operationally trivial sector)**.** For each accepted tick k, choose a continuous symmetric coercive bilinear form

$$\langle\cdot,\cdot\rangle^{\mathrm{aux}}_{\Lambda_k} : \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \times \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proposition 1.17 (Strong-package calibration extension from active stiffness)**.** *Assume assumptions 1.1 to 1.3. Set*

$$\chi_{\Lambda_k} := \rho_{\Lambda_k}, \qquad \mathcal{T}_{\Lambda_k} := \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Then the formula

$$\mathsf{B}_{\Lambda_k}(F,G) := \widetilde{\alpha}(\Lambda_k)\,\chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \big\langle F - \chi_{\Lambda_k}(F)J^{\mathrm{act}}_{\Lambda_k},\, G - \chi_{\Lambda_k}(G)J^{\mathrm{act}}_{\Lambda_k}\big\rangle^{\mathrm{aux}}_{\Lambda_k} \tag{5}$$

defines a continuous symmetric positive definite bilinear form on $E^-_{\Lambda_k}$ satisfying

$$\mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, J^{\mathrm{act}}_{\Lambda_k}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, T) = 0 \quad \forall T \in \mathcal{T}_{\Lambda_k}.$$

Proof. By proposition 1.10, every $F \in E^-_{\Lambda_k}$ has a unique decomposition

$$F = \chi_{\Lambda_k}(F)\, J^{\mathrm{act}}_{\Lambda_k} + T_F, \qquad T_F \in \mathcal{T}_{\Lambda_k}.$$

Equation (5) is therefore well-defined, continuous, and symmetric. Positivity is immediate from $\widetilde{\alpha}(\Lambda_k) > 0$ together with coercivity of $\langle\cdot,\cdot\rangle^{\mathrm{aux}}_{\Lambda_k}$ on $\mathcal{T}_{\Lambda_k}$. If $\mathsf{B}_{\Lambda_k}(F,F) = 0$, then both $\chi_{\Lambda_k}(F) = 0$ and $\langle T_F, T_F\rangle^{\mathrm{aux}}_{\Lambda_k} = 0$, hence $T_F = 0$ and $F = 0$. The displayed normalization and orthogonality properties follow by substitution into (5). □

Corollary 1.18 (Descendant ambiguity compatibility on the characterization layer)**.** *For each accepted tick k, the odd-projected ambiguity directions coming from*

$$Q_t X + Q_u Y, \qquad \partial_\mu K^\mu, \qquad \Delta_{\mathrm{ct}}^-$$

belong to $\mathcal{T}_{\Lambda_k}^{\mathrm{op}}$ whenever K^μ is taken from the boundary-admissible class of proposition 1.13 and Δ_{ct}^- is admissible in the sense of definition 1.14.

Proof. Combine propositions 1.12, 1.13 and 1.15. □

Theorem 1.19 (Strong rank-one realization of the characterization layer)**.** *Assume assumptions 1.1 to 1.4, 1.6 and 1.7. Then for each accepted tick k, the split/leak construction exports the UOM-side data*

$$\big(a_{\Lambda_k},\, S_{\Lambda_k},\, J_{\Lambda_k}^{\mathrm{act}},\, C_{\Lambda_k}^{\mathrm{act}},\, \widetilde{\alpha}(\Lambda_k),\, \rho_{\Lambda_k}\big), \tag{6}$$

where:

1. *$\mathcal{O}_{\Lambda_k}^{\mathrm{char}} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ is the minimal completion-sector characterization algebra;*
2. *the operationally trivial subspace satisfies*
$$\mathcal{T}_{\Lambda_k} := \mathcal{T}_{\Lambda_k}^{\mathrm{op}} = \ker R_{\Lambda_k}^{\mathrm{phys}} = \ker \rho_{\Lambda_k};$$
3. *the intrinsic physical response is one-dimensional on the strong sector:*
$$R_{\Lambda_k}^{\mathrm{phys}}(F) = \rho_{\Lambda_k}(F)\, C_{\Lambda_k}^{\mathrm{act}}, \qquad R_{\Lambda_k}^{\mathrm{phys}}(J_{\Lambda_k}^{\mathrm{act}}) = C_{\Lambda_k}^{\mathrm{act}} \neq 0;$$
4. *the active odd quotient is rank one and is generated by $[J_{\Lambda_k}^{\mathrm{act}}]$;*
5. *the abstract orientation functional of the transported interface is realized by*
$$\chi_{\Lambda_k} := \rho_{\Lambda_k};$$
6. *any admissible calibration form B_{Λ_k} in this strong package may be obtained by extending the active stiffness $\widetilde{\alpha}(\Lambda_k)$ with proposition 1.17.*

Proof. Definition 1.1 packages the active completion-sector channel extracted from the split/leak construction: a_{Λ_k} is the distinguished odd split coordinate singled out by sections 1.1 and 1.3, while $S_{\Lambda_k} = a_{\Lambda_k}^2$ is the even stock entering (16) and (19). The active odd generator $J_{\Lambda_k}^{\mathrm{act}}$, active physical covariance tangent $C_{\Lambda_k}^{\mathrm{act}}$, and scalar response coefficient ρ_{Λ_k} are exported by assumption 1.1; proposition 1.10 and corollary 1.11 then identify the abstract interface-trivial subspace and the corresponding rank-one quotient. The split/leak law fixes the active stiffness $\widetilde{\alpha}(\Lambda_k)$, and proposition 1.17 upgrades that one-line datum to an admissible full calibration form once the auxiliary coercive core is chosen. Finally, corollary 1.18 places the odd-projected descendant ambiguities in $\mathcal{T}_{\Lambda_k}$, yielding the stated ambiguity compatibility on the characterization layer. □

Theorem 1.20 (Intrinsic weak transported descendant-class interface)**.** *Assume the hypotheses of **??** on a connected accepted-branch segment I. Then for every $\Lambda \in I$, the sufficiently local odd generator layer carries the weak transported interface data*

$$\big(E_\Lambda^-,\, \mathcal{A}_\Lambda^-,\, [\Xi_\Lambda^{(1)}],\, \chi_\Lambda,\, \varepsilon_\Lambda\big)$$

with

$$\chi_\Lambda := \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}},$$

and the following conclusions hold.

1. *there exists a representative*
$$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}]$$
such that
$$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) \neq 0;$$
equivalently, the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;

2. *the orientation functional is realized intrinsically on the sufficiently local generator layer:*
$$\mathcal{A}_\Lambda^- \subset \ker \chi_\Lambda, \qquad \chi_\Lambda(J) = \varepsilon_\Lambda \quad \forall J \in [\Xi_\Lambda^{(1)}];$$

3. *along I, the sign ε_Λ is locally constant.*

In particular, the weak transported interface is realized intrinsically, without appeal to completion coordinates, under the stated hypotheses.

Proof. The functional $\rho_\Lambda^{\mathrm{dom}}$ is defined intrinsically from the sufficiently local generator layer by definition 1.3 together with the dominant-line prescription of **??**. Item (1) is exactly the nonzero-overlap conclusion of item (3) in **??**, item (2) is exactly item (4) there, and item (3) is item (5). □

Theorem 1.21 (Quadratic-probe realization of the weak activity supplement)**.** *Assume the hypotheses of* **??** *on a connected accepted-branch segment I. For each $\Lambda \in I$, define the even quadratic probe sector by*
$$\mathcal{P}_\Lambda^+ := \mathrm{span}\{\mathcal{Q}_{f,g}^\Lambda : f, g \text{ admissible band-limited receiver probes}\} \subset E_\Lambda^+.$$
Then, for every $F \in E_\Lambda^-$,
$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+$$
implies
$$\chi_\Lambda(F) = 0.$$
Equivalently, if $\chi_\Lambda(F) \neq 0$, then there exists a sufficiently local even quadratic probe $A \in \mathcal{P}_\Lambda^+$ such that
$$\{F, A\}_{\mathrm{P},\Lambda} \neq 0.$$
In particular, the weak interface data of theorem 1.20 carry a canonical activity witness sector built from the intrinsic quadratic receiver probes.

Proof. Assume
$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+.$$
In particular,
$$\{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} = 0$$
for every admissible pair f, g. By definition 1.3, this means
$$\langle f, \mathcal{R}_\Lambda(F) g\rangle = \left\langle \{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} \right\rangle_{\rho_{*,\Lambda}} = 0$$
for all admissible f, g, hence $\mathcal{R}_\Lambda(F) = 0$. Therefore
$$\Pi_{\mathrm{phys},\Lambda} \mathcal{R}_\Lambda(F) = 0,$$

and the definition of the raw dominant-response coefficient in **??** gives

$$\widetilde{\rho}_{\Lambda}^{\mathrm{dom}}(F) = 0.$$

By item (1) of **??**, $\delta C_{\Lambda}^{\mathrm{LoG}} \neq 0$, so dominant-line realization holds and $\rho_{\Lambda}^{\mathrm{dom}}$ is well defined. Hence

$$\rho_{\Lambda}^{\mathrm{dom}}(F) = 0.$$

Now apply theorem 1.20, which identifies $\chi_{\Lambda} = \nu_{\Lambda}^{-1}\rho_{\Lambda}^{\mathrm{dom}}$, to obtain

$$\chi_{\Lambda}(F) = 0.$$

The contrapositive gives the equivalent detection statement. □

Remark 1.22 (Correct scalar form of representative comparisons). The weak export theorem should be read ρ^{dom}-first on the ambiguity quotient. Because

$$\chi_{\Lambda} := \nu_{\Lambda}^{-1}\rho_{\Lambda}^{\mathrm{dom}},$$

any comparison statement that depends only on dominant pairing is naturally a statement about the descended scalar

$$\overline{\rho}_{\Lambda}^{\mathrm{dom}} : E_{\Lambda}^{-}/\mathcal{A}_{\Lambda}^{-} \to \mathbb{R}$$

or, equivalently, about its value on the transported descendant affine class. Such a statement may be rewritten in χ_{Λ} form only when the normalization is made explicit; for example, if $\nu_{\Lambda} = 1$, then the same scalar value is represented by a descended $\bar{\chi}_{\Lambda}$. Without that normalization the stronger unscaled χ-compatibility form is false in general.

Corollary 1.23 (UOM-stiffness specialization of the weak transported calibration)**.** *Assume assumption 1.7 and fix an accepted tick k for which the conclusions of theorem 1.20 hold. Suppose* $\ker \chi_{\Lambda_k}$ *admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it, and choose any*

$$J_{\Lambda_k}^{\sharp} \in E_{\Lambda_k}^{-} \qquad \textit{with} \qquad \chi_{\Lambda_k}(J_{\Lambda_k}^{\sharp}) = 1.$$

Then **??** *may be applied with*

$$\beta_{\Lambda_k} := \widetilde{\alpha}(\Lambda_k)$$

to produce a continuous symmetric positive definite bilinear form

$$\mathsf{B}_{\Lambda_k}^{\widetilde{\alpha}} : E_{\Lambda_k}^{-} \times E_{\Lambda_k}^{-} \to \mathbb{R}$$

such that

$$\mathsf{B}_{\Lambda_k}^{\widetilde{\alpha}}(J_{\Lambda_k}^{\sharp}, J_{\Lambda_k}^{\sharp}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}_{\Lambda_k}^{\widetilde{\alpha}}(J_{\Lambda_k}^{\sharp}, K) = 0 \quad \forall K \in \ker \chi_{\Lambda_k}.$$

In particular,

$$\mathcal{A}_{\Lambda_k}^{-} \subset \ker \chi_{\Lambda_k}$$

implies

$$\mathsf{B}_{\Lambda_k}^{\widetilde{\alpha}}(J_{\Lambda_k}^{\sharp}, A) = 0 \qquad \forall A \in \mathcal{A}_{\Lambda_k}^{-}.$$

Proof. By remark 1.16, the split/leak package fixes the positive stiffness scale $\widetilde{\alpha}(\Lambda_k)$, and assumption 1.7 records its positivity and asymptotic size. Apply **??** with $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$. The final orthogonality statement follows from item (2) of theorem 1.20. □

Remark 1.24 (Inherited ledger compatibility). The inherited commutative ledger algebra is intentionally not part of the core characterization algebra $\mathcal{O}_{\Lambda_k}^{\mathrm{char}}$. Its role is separate: the split model requires the active odd representative to be completion-sector nontrivial while acting trivially, to leading order, on the inherited classical ledger. This is the UOM-side compatibility statement behind the slogan "oddness in the completion sector, classicality in the accessible ledger."